Jason Yang

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SUMMARY

Master of Software Development student at Victoria University of Wellington with prior engineering R&D experience and a background in securities operations support. Hands-on software projects in backend APIs and integration tooling. Comfortable building, testing, and debugging systems end-to-end, with strong documentation and cross-team communication. Seeking a Software Engineering Internship where I can contribute quickly and keep learning.

EDUCATION

Master of Software Development — Victoria University of Wellington — Jul 2025–Jun 2026 (Expected)

Master of Power Engineering & Engineering Thermophysics — Jiangsu University — 2015–2018

BEng Thermal Energy & Power Engineering — China University of Mining and Technology — 2010–2014

PROJECTS

BookExpress Assistant Client — Desktop Validation & Audit Tool (Shopify · Omnivore · TradeMe)
(Team; owned SQLite migration + audit persistence)

- Built a desktop client to validate listing rules and support controlled submissions across Shopify → Omnivore → TradeMe, improving data correctness and traceability.
- Added embedded SQLite persistence with automatic schema migration on startup (create/alter + seed).
- Ensured audit integrity with a transaction wrapper around sync_record + detail writes, preventing partial states and powering Records history.
- Investigated and fixed a metafield namespace mismatch affecting listing condition values; documented repro steps and verification.

Bank Account Service API (Java 17, Spring Boot, JPA/Hibernate, PostgreSQL/H2)

- Built a Spring Boot banking API (accounts, deposit/withdraw, transfers) enforcing business rules in the service layer.
- Implemented transactional money transfers to update two accounts atomically and prevent invalid balance states (e.g., overdraft).
- Persisted immutable transaction history (audit trail) via JPA/Hibernate; exposed REST endpoints with Swagger UI; verified logic with unit tests.

Meeting Room Reservation API (C#, ASP.NET Core, EF Core, SQLite, Swagger/OpenAPI)

- Built a room booking REST API with layered architecture (Controller → Service → Domain → EF Core/SQLite).

- Implemented server-side time-range conflict detection to prevent double-booking and enforce consistent rules across clients.
- Integrated EF Core + SQLite persistence and Swagger/OpenAPI; verified endpoints via Swagger UI/Postman and documented DTOs.

Time-Series Backtesting Demo (Python, pandas, NumPy)

- Built a config-driven backtesting pipeline (Data → Signals → Execution → Metrics → Reports) on historical OHLCV data.
- Prevented look-ahead bias by separating signal generation from execution; enforced t+1 open execution with transaction cost modeling.
- Generated performance/risk outputs (equity curve, drawdown, Sharpe/volatility) and ensured reproducibility via dataset/config hashing.

EXPERIENCE

Investment Operations & Strategy Support — Huaxi Securities / Shanxi Securities — Apr 2022–Jan 2024

- Analysed structured financial/operational data to support strategy evaluation and reporting.
- Automated public data collection and processing using Python scripts; improved repeatability through validation checks.
- Coordinated workflows across trading, custody, and operations teams; communicated findings to internal stakeholders.

R&D Engineer — BOE Technology Group — Jul 2018–Jan 2021

- Investigated production issues by analysing test results and process data to identify root causes and corrective actions.
- Worked with equipment/process/quality teams to resolve system-level faults and restore normal operation.
- Standardised testing and reporting procedures to improve repeatability and traceability; documented fixes for internal reference.

TECHNICAL SKILLS

Languages: Java, C#, Python, SQL

Backend & APIs: Spring Boot, ASP.NET Core, REST API design, validation, layered architecture, debugging

Databases/ORM: PostgreSQL, SQLite, JPA/Hibernate, EF Core

Testing/Tools: Git, JUnit, Swagger/OpenAPI, Postman, API testing, documentation

Frontend: JavaScript, HTML, CSS (basic React)

INTERESTS

Hiking · Tennis · Billiards · Films · Video Games